MAPS FROM A QUASI-COMPACT ALGEBRAIC SPACE INTO A COLIMIT OF THICKENINGS FACTOR THROUGH ONE TERM

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The delivered context span is the author's own definition of an affine formal algebraic space, which the statement invokes; it is copied whole.

Editorial note. No bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The presentation X = colim X_lambda of an affine formal algebraic space and the discussion following it. Same chapter, Definition definition-affine-formal-algebraic-space and the discussion after it, invoked at proof lines 2081-2082 and 2090-2091. Provides the terms X_lambda and the factoring statement for morphisms from affine schemes.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is an *affine formal algebraic space* if there exist

(1) a directed set Λ,
(2) a system $(X_\lambda, f_{\lambda\mu})$ over Λ in $(Sch/S)_{fppf}$ where
   (a) each X_λ is affine,
   (b) each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening,

such that

$$X \cong \operatorname{colim}_{\lambda\in\Lambda} X_\lambda$$

as fppf sheaves and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of affine formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let $X_\lambda, \lambda \in \Lambda$ and $X = \operatorname{colim} X_\lambda$ be as in Definition 0.1. If Y is a quasi-compact algebraic space over S, then any morphism $Y \to X$ factors through an X_λ.*

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Proof. Choose an affine scheme V and a surjective étale morphism $V \to Y$. The composition $V \to Y \to X$ factors through X_λ for some λ by the discussion following Definition 0.1. Since $V \to Y$ is a surjection of sheaves, we conclude. $\square$